

Om Sanjay Singhan

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EDUCATION

NEW YORK UNIVERSITY

Master of Science (MS), Computer Engineering — GPA: 3.89/4 · Dean's List

New York, NY
September 2023 – May 2025

NMIMS (DEEMED-TO-BE UNIVERSITY)

Bachelor of Technology in Computer Engineering — GPA: 3.76/4

Mumbai, India
July 2019 – May 2023

SKILLS

- **Programming Languages:** Java, TypeScript, C#, JavaScript, Python, C/C++, Bash, R, SQL
- **Frontend Frameworks:** React, Redux, HTML5, CSS, Tailwind CSS, D3.js
- **Backend Frameworks:** GraphQL, ASP.NET, Node.js, Fastify, Express, PostgreSQL, MySQL, NoSQL, MongoDB, DynamoDB
- **Cloud & DevOps:** AWS, Azure, Docker, Kubernetes, Git, Jenkins, CI/CD, Linux, CMake
- **ML Frameworks:** PyTorch, TensorFlow, Keras, NumPy, Pandas, NLTK
- **Agentic AI & LLMs:** GPT-4, LangChain, Agentic Workflows, Prompt Engineering, RAG, MCP Servers, Vector Retrieval
- **Practices:** RESTful API Design, Microservices Architecture, Agile/Scrum, Test-Driven Development

PROFESSIONAL EXPERIENCE

Sewer AI

Software Engineer (Full-time)

San Francisco, CA
September 2025 – Present

- **Full-Stack API Development:** Shipped and maintained secure, production RESTful APIs and backend services in Node.js, TypeScript, Fastify, and PostgreSQL, applying algorithmic and architectural optimizations to sustain 10,000+ requests with sub-500ms latency.
- **Test-Driven Development:** Strengthened engineering craft and reliability through Jest unit/integration and Playwright end-to-end testing, reducing production bugs by 40% and achieving 85% code coverage.
- **Microservices Architecture:** Owned fault-tolerant, state-aware distributed pipelines (Temporal.io) end-to-end, from design through production operation, processing 10,000+ hours of inspection data daily at 99.5% reliability.
- **Incident Response & Triage:** Operated and monitored production systems during on-call rotations, triaging incidents and applying root cause analysis to restore service reliability and minimize downtime.
- **Collaborative Development:** Drove team knowledge-sharing through Git-based version control and peer code reviews within Agile/Scrum sprint cycles; authored technical documentation for backend services and data pipelines to support long-term maintainability and onboarding.

MAK Capital

Artificial Intelligence (AI) Developer (Intern)

New York, NY
February 2025 – May 2025

- **Multimodal RAG Integration:** Engineered and shipped a multimodal RAG pipeline (Azure Document Intelligence, GPT-4, LangChain), boosting response accuracy 40% through semantic chunking and text/image analysis.
- **ETL Pipeline Design:** Architected scalable cloud data ingestion, transformation, and storage pipelines (Python, AWS Lambda, S3), converting raw JSON into governed datasets for analytics and exploratory analysis.
- **Automation & Predictive Modeling:** Developed an automated sentiment analysis system (BeautifulSoup, Scrapy, FinBERT) to identify short opportunities from financial news, delivering 80% prediction accuracy and trading alpha.

Aroris Health

Software Development Engineer (Intern)

Minneapolis, MN
May 2024 – August 2024

- **Secure, Scalable System Design:** Designed secure, scalable API infrastructure (Python, Firebase) with HIPAA-compliant authentication, protecting 1,000+ patient health records while building data governance systems for AI-ready pipelines.
- **Cross-Functional Product Collaboration:** Partnered with product managers and UX designers to translate user needs into data-centric frontend features for analytics and reporting workflows, applying core UI/UX principles to improve usability.
- **Security-First Design:** Implemented OAuth 2.0 authentication, SQL injection prevention, and encrypted data transmission following OWASP guidelines to protect sensitive patient data.
- **Monitoring & Observability:** Operated and continuously improved system observability by building and maintaining monitoring dashboards and alerting rules, enabling faster detection and resolution of pipeline and API issues.

LEADERSHIP EXPERIENCE

Research Assistant, New York University

- **Mentorship & Code Review:** Mentored 2 junior developers on code review practices and software design principles.

ACADEMIC PROJECTS

NSF Funded Project: Distribution System Situational Awareness via Continuous-Time Adaptive Data Fusion

New York University — Generative AI, TimeScaleDB, Express, Python, TensorFlow

- **AI Simulation & Model Lifecycle:** Designed and trained a power grid simulator (TensorFlow) generating 150,000+ synthetic samples at 88% accuracy, with feedback loops for iterative refinement and drift resilience.
- **Time-Series Analytics & Visualization:** Delivered user-facing D3.js dashboards for real-time and historical metrics, improving query performance 50% via chart-selection and color-hierarchy best practices.